Version: 01

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Centenary Celebrated Sharnbasveshwar Vidya Vardhaka Sangha's



ಶರಣಬಸವ Sharnbasva ವಿಶ್ವವಿದ್ಯಾಲಯ University

ಕಲಬುರಗಿ–585 103
ಕರ್ನಾಟಕ–ಭಾರತ
ಸ್ಥಾಪನೆ : 2017

Kalaburagi-585 103
Karnataka-India
Estd.: 2017



Faculty of: Engineering & Technology	Department: CSE/CSD Program: B. Tech	AY:2026-2027

INTERNAL ASSESSMENT (I.A.) TEST: I

Course Title: Computer Organization & Architecture	Course Code: 25CS303
Date: 04-11-2026	Time: 10.00AM-11.30AM
Semester and Section: III	Max. Marks:30

Note: Answer any three full questions selecting at least one question from each module

Q. No.		QUESTIONS	Marks	CO's	RBT Levels
		MODULE-I			
1	a)	Explain the functional units of a computer with a neat block diagram.	06	CO1	L2
	b)	Apply the basic operational concepts to explain how data is transferred between memory and Processor using given instruction Load LOCA,R1 Add R1,R0	06	CO1	L3
		OR			
2	a)	Apply signed binary representation techniques to represent the decimal values 5, −2, 14, −10, 51, and −43 using 8-bit formats: (a) Sign-and-Magnitude, (b) 1's Complement, and (c) 2's Complement.	06	CO1	L2
	b)	Identify the different Addressing modes with suitable example	06	CO1	L3
		MODULE-II			
3	a)	Apply stack operations to determine the final contents of the stack for a given sequence of PUSH and POP instructions.	06	CO2	L3
	b)	Describe subroutine linkage and the sequence of operations involved in calling and returning from a subroutine.	06	CO2	L3
		OR			
4	a)	Apply the basic methods of accessing I/O devices	06	CO2	L3
	b)	Apply the sequence of events involved in interrupt handling during an I/O operation.	06	CO2	L3
		MODULE-III			
5	a)	Apply the concept of DMA to explain how data can be transferred between an I/O device and memory without continuous CPU involvement.	06	CO3	L3
		OR			
6	a)	Analyze the internal organization and operation of a typical SRAM memory cell. Explain how data is read from and written into SRAM.	06	CO3	L4

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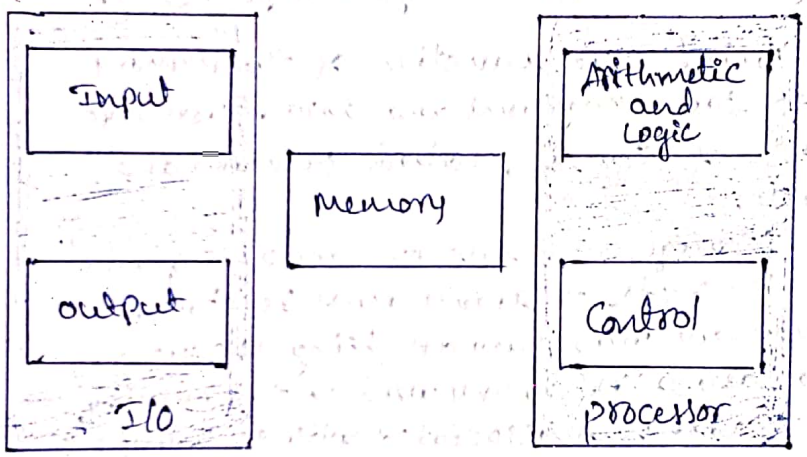

Marks Distribution of CO's and RBT Level wise						
CO's	L1	L2	L3	L4	Total Marks	% of Marks Allotted
CO1	0	12	12	0	24	40%
CO2	0	0	24	0	24	40%
CO3	0	0	06	06	12	20%
Total RBT Marks	0	12	42	06	60	100%
% of RBT Levels	0%	20%	70%	10%	100%	

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Faculty of: Engineering and Technology	Department: CSD. Program: B.Tech.	AY: 2026-2027

INTERNAL ASSESSMENT (I.A.) TEST: I

SCHEME OF EVALUATION	
Course Title: Computer Organization and Architecture	Course Code: 25CS303.
Semester: III	Section:
Course Teacher Prof. Santosh.	Contact Number: 8050204767

Q.No.	Answer description	Marks Allotted
1 a)	 Block diagram of Basic functional units of a Computer. * A Computer consists of five functionally independent main parts: input, memory, arithmetic and logic, output, and control units. The input unit accepts coded information from human operators, from electromechanical devices such as key boards or mouse. whenever a key is pressed, the corresponding letter or digit is automatically translated into its corresponding binary code and transmitted over a cable to either the	06M 3M

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Q.No.	Answer description	Marks Allotted
	Memory or the processor. Many other kinds of input devices are available; including joysticks, trackballs and mouses. These are often used as graphic input devices in conjection with displays. Microphones can be used to capture audio input which is then sampled and converted into digital codes for storage and processing. Memory Unit:- The function of the memory unit is to store programs and data. There are two classes of storage, called primary and secondary. Primary storage is a fast memory that operates at electronic speeds. programs must be stored in the memory while they are being executed. To provide easy access to any word in the memory, a distinct address is associated with each word location. Secondary storage is used when large amounts of data and many programs have to be stored particularly for information that is accessed infrequently. Arithmetic and logic Unit:- Most computer operations are executed in the arithmetic and logic unit (ALU) of the processor. example: Suppose two numbers located in the memory are to be added. They are brought into the processor, and the actual addition is carried out by the ALU. The sum may then be stored in the memory or retained in the processor for immediate use. Example: Addition, Multiplication, division or comparison of numbers.	03 M

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Q.No.	Answer description	Marks Allotted
	Output Unit: The output unit is the counterpart of the input unit. Its function is to send processed results to the outside world. The most familiar example of such a device is a printer. Control Unit: The memory, ALU and input and output units store and process information and perform input and output operations. These units must be coordinated in some way. The control unit is effectively the nerve center that sends control signals to other units and senses their states.	
1 b)	Load LOCA, R_1 Add R_1, R_0 Data transfer between the processor (CPU) and main memory is performed using instructions. The CPU executes instructions with the help of registers, ALU and the control unit. Example:- * LOCA is a memory location * R_1 and R_0 are CPU registers * The contents of memory location LOCA are 20 * The initial contents of register R_0 are 10 Load LOCA, R_1 This instruction transfer data from memory location LOCA into register R_1. $R_1 \leftarrow M[LOCA]$	6M Diagram

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Q.No.	Answer description	Marks Allotted
	* The CPU fetches the LOAD instruction from memory. * The control unit decodes the instruction. * The address of memory location LOCA is placed in the memory address register (MAR) * The control unit generates a memory read signal. * Data (20) is transferred from memory to the memory data register (MDR) * The data is copied from MDR into register R_1. $R_1 = 20$ ADD R_1, R_0 This instruction adds the contents of registers R_1 and R_0 and stores the result in R_0. $R_0 \leftarrow R_1 + R_0$ * The CPU fetches and decodes the ADD instruction * The contents of R_1 (20) and R_0 (10) are sent to the ALU. * ALU performs the addition operation * The result (30) is stored in register R_0. * Status flags may be updated according to the result. $R_0 = 20 + 10 = 30$ * Register values After Execution Register/Memory Initial value Final value LOCA 20 20 R_1 0 20 R_0 10 30	03M 03M

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Q.No.	Answer description	Marks Allotted
2 a)	Signed binary representation is used to represent both positive and negative numbers in a computer system. In an 8-bit representation, the first bit represents the sign and the remaining bits represent the value (magnitude).	06

Converting the given decimal numbers into 8-bit binary

Decimal	8-bit binary	Magnitude	sign
+5	00000101	00000101	
−2	00000010	10000010	
14	00001110	00001110	
−10	00001010	10001010	
51	00110011	00110011	
−43	00101011	10101011	

Decimal	sign and magnitude	1's Complement	2's Complement	
+5	00000101	00000101	00000101	— 01M
−2	10000010	11111101	11111110	— 01M
+14	00001110	00001110	00001110	— 01M
−10	10001010	11110101	11110110	— 01M
51	00110011	00110011	00110011	— 01M
−43	10101011	11010100	11010101	— 01M

* 1's Complement representation
- positive numbers remain unchanged.
- Negative numbers, invert all bits (0 becomes 1 and 1 becomes 0)

* 2's Complement representation.
- positive numbers unchanged.
- Negative numbers, first find 1's complement then add 1.

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Q.No.	Answer description	Marks Allotted
2 b)	Addressing modes are the methods used by the CPU to specify the location of an operand (data) required to execute an instruction. The different types of addressing modes are explained below. ① Immediate Addressing Mode:- * In this mode, the actual data (operand) is given directly in the instruction. * No additional memory access is required to obtain the operand. Ex:- MOV R1, #25; The value 25 is directly loaded into register R1. ② Direct Addressing Mode:- * In this mode, the memory address of the operand is specified directly in the instruction. * The CPU accesses the data from that memory location. LOAD R1, 5000; The data stored at memory address 5000 is loaded into register R1. ③ Indirect Addressing Mode:- * In this mode, the instruction specifies a memory location that contains the actual address of the operand. * The CPU first accesses the address and then retrieves the data. LOAD R1, (5000); If memory location 5000 contains address 6000, the CPU loads the data stored at address 6000 into R1.	06 Mentioning Names 03 M Examples 03 M

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Q.No.	Answer description	Marks Allotted

④ Register Addressing Mode:-
* In this Mode, the operand is stored in a CPU register.
* ADD R_1, R_2; The contents of register R_2 are added to register R_1.

⑤ Register Indirect Addressing Mode:-
* In this Mode, a register contains the Memory address of the operand.
* The CPU uses the address stored in the register to access data from memory.
LOAD R_1, (R_2); if R_2 Contains address 4000, the data stored at memory location 4000 is loaded into R_1.

⑥ Indexed Addressing Mode:-
* In this Mode, the effective address is calculated by adding an index register to the base address.
* It is commonly used to access array elements.
LOAD R_1, 1000(R_2); if R_2 Contains 20, the effective address is EA = 1000+20=1020
The data at address 1020 is loaded into R_1.

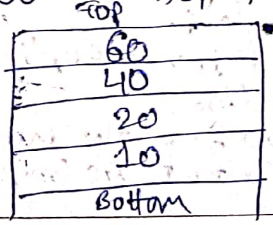
⑦ Relative Addressing Mode:-
* In this Mode, the effective address is calculated by adding an offset to the program Counter (PC).
* It is commonly used in Branch instructions.
BEQ +20; if PC Contains 1000 the branch target address is 1020 assuming the offset is measured in bytes.

⑧ Implicit Addressing Mode:-
* In this Mode, the operand is not Explicitly mentioned in the instruction.
* The operand is understood from the instruction itself.
CMA ; The accumulator Contents are Complemented.

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Q.No.	Answer description	Marks Allotted
3 a)	A stack is a linear data structure that follows the LIFO (Last In, First out) - this means the elements inserted last will be removed first. PUSH :- Inserts an element into the top of the stack. POP :- Removes the top element from the stack. The stack uses a pointer called TOP, which indicates the position of the topmost element. * Consider the following sequence of PUSH and POP operations: PUSH 10 PUSH 20 PUSH 30 POP PUSH 40 PUSH 50 POP PUSH 60 Example:-	06 M 03 M

Step	Instruction	Stack Contents (Bottom to Top)	Top
1	PUSH 10	10	10
2	PUSH 20	10, 20	20
3	PUSH 30	10, 20, 30	30
4	POP	10, 20	20
5	PUSH 40	10, 20, 40	40
6	PUSH 50	10, 20, 40, 50	50
7	POP	10, 20, 40	40
8	PUSH 60	10, 20, 40, 60	60



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Q.No.	Answer description	Marks Allotted
3 b)	Subroutine linkage is the mechanism used by the CPU to transfer control from the main program to subroutine and return control back to the main program after completing the task. It helps in code reusability, reduces program size and avoids writing the same instructions repeatedly. It involves two important operations. 1) Subroutine Calling:- Transfers control from the main program to the subroutine. 2) Subroutine Returning:- Transfers control from the subroutine back to the main program. Program Counter (PC) to store the address of the next instruction to be executed. The return address is saved in a register or stack so that execution can continue from the correct location. * The main program executes a CALL instruction. * The CPU identifies the starting address of the subroutine. * The address of the next instruction (return address) is saved in the stack or a link register. * The PC is loaded with the starting address of the subroutine. * The CPU starts executing the subroutine instructions. Example:- MAIN: MOV R1, 10 CALL ADDITION MOV R2, 20	06 M 02 M

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Q.No.	Answer description	Marks Allotted

STOP

ADDITION:

ADD R1, R1

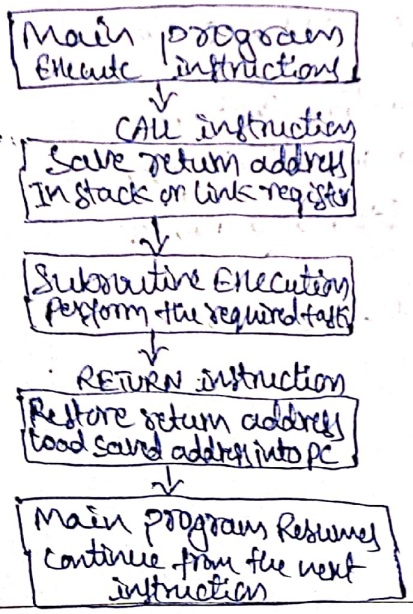
RETURN

- The main program loads 10 into register R1
- CALL ADDITION transfers control to the subroutine
- The return address is saved.
- The subroutine executes ADD R1, R1.
- RETURN transfers control back to the main program.
- The main program continues with the next instruction.

02M

When the subroutine completes its execution

- The subroutine completes all its required instructions
- The CPU executes the RETURN instruction.
- The saved return address is retrieved from the stack or link register.
- The return address is loaded into the PC.
- Control returns to the main program.
- The main program continues execution from the instruction immediately following the CALL instruction.

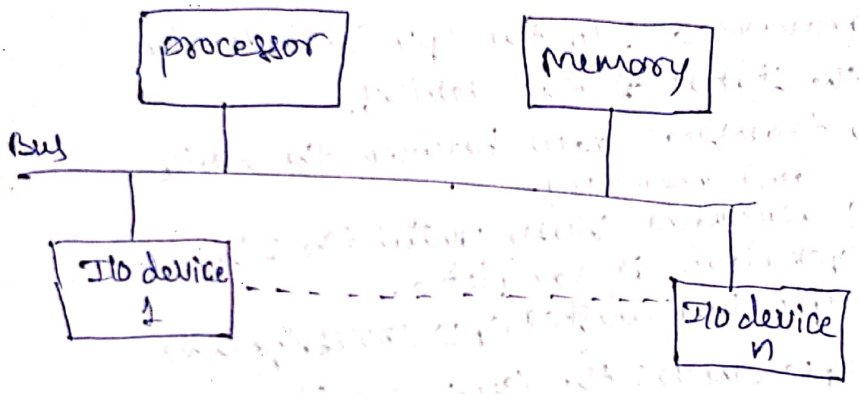


02M

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Q.No.	Answer description	Marks Allotted
4 a)	The bus enables all the I/O devices connected to it to exchange information. It consists of three sets of lines used to carry address, data and control signals. Each I/O device is assigned a unique set of addresses. Ex:- Keyboards, printers, monitors & storage devices The CPU uses three basic methods to access I/O devices. 1) programmed I/O :- • In programmed I/O the CPU continuously checks the status of the devices. • The CPU transfers data between the device and memory. • The CPU remains busy until the I/O operation is completed. • It is simple but wastes CPU processing time. Ex:- The CPU checks the keyboard status and reads the character when a key is pressed. 2) Interrupt-driven I/O :- • In this type, the I/O device is ready, it sends an interrupt signal to the CPU. • The CPU temporarily stops its current task and executes the Interrupt Service Routine (ISR) • After completing the operation the CPU resumes its previous task. Ex:- When a keyboard key is pressed, the keyboard generates an interrupt to inform the CPU. 3) Direct Memory Access (DMA) :- • DMA allows data transfer directly between an I/O device and main memory.	06 M 2 M 04 M

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Q.No.	Answer description	Marks Allotted
	• The CPU initializes the DMA Controller by providing the starting address, data count and direction of transfer. • The DMA Controller manages the data transfer without continuous CPU involvement. • After completing the transfer, DMA sends an interrupt to the CPU. Ex:. Transferring a large file from a hard disk to RAM using DMA. 	02 M
4 b)	I/O Device Generates Interrupt ↓ CPU Completes Current Instruction ↓ Saves CPU state ↓ Identifies Interrupt and Executes ISR ↓ Clears Interrupt Requests ↓ Restores CPU State ↓ Returns Previous Program	06 M 03 M

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Q.No.	Answer description	Marks Allotted
	Interrupt handling is a process in which an I/O device sends an interrupt signal to the CPU to request its attention. The CPU temporarily stops its current execution handles the interrupt, and then resumes the previous task. Sequence of Events Involved in Interrupt Handling. 1) I/O Device Generates an Interrupt. • When an I/O device requires attention, it sends an interrupt signal to the CPU. • EX: A keyboard generates an interrupt when a key is pressed. 2) CPU Completes the Current Instruction. • The CPU completes the instruction currently being executed. • It then checks for pending interrupt requests. 3) CPU saves the Current State • The CPU saves the contents of the program Counter (PC) and necessary processor registers. • This helps the CPU resume the previous program later. 4) CPU Identifies the Interrupt • The CPU identifies which I/O device generated the interrupt. • It obtains the address of the appropriate Interrupt Service Routine (ISR). 5) Executes the Interrupt Service Routine. • The CPU executes the ISR to service the I/O device. • Ex: It reads the data received from the keyboard.	3 M

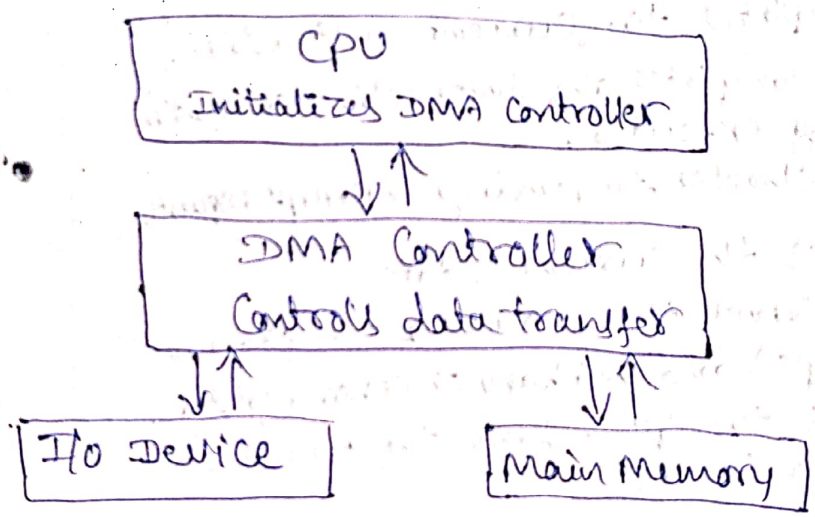
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Q.No.	Answer description	Marks Allotted

6) Clears the Interrupt
- After completing the I/O operation, the CPU sends an acknowledgement to the device.
- The interrupt request is cleared.

7) Restores the previous state.
- The CPU restores the saved registers and program counter.
- It resumes execution of the interrupted program.

5 a) Block Diagram of DMA — 06 M



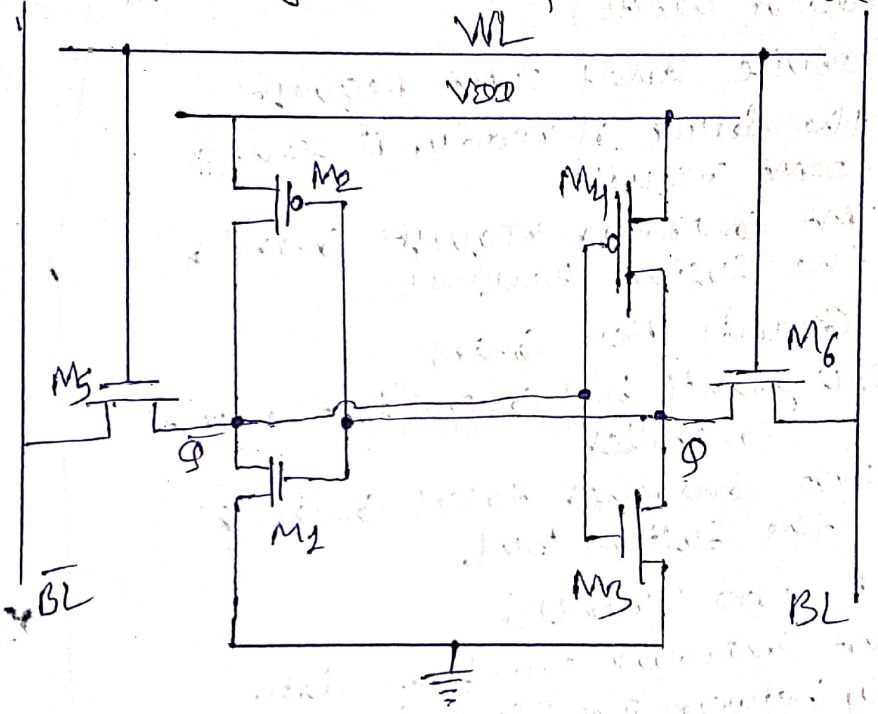
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Direct Memory Access (DMA) is a data transfer technique that allows an I/O device to transfer data directly to or from main memory without requiring continuous involvement of the CPU. This reduces the workload of the CPU and improves system performance.

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Q.No.	Answer description	Marks Allotted
	Working of DMA :-	03 M
	1) CPU Initializes DMA Controller • The CPU provides the starting memory address. • It specifies the number of bytes or words to be transferred. • It specifies the direction of data transfer (Read or write) 2) I/O Device Sends DMA Request • When the device is ready, it sends a DMA request. • The DMA Controller requests control of the system bus. 3) CPU Grants Bus Control • The CPU grants bus control to the DMA Controller • The DMA Controller takes control of the system bus. 4) Direct Data Transfer • The DMA Controller transfers data directly between the I/O device and main memory. • The CPU can perform other tasks during this transfer, subject to bus availability. 5) Update Address and Count • After each transfer the DMA Controller updates the memory address. • It decreases the remaining data count. 6) Completion of Transfer • When all data is transferred, the DMA Controller releases the bus. • It sends an interrupt signal to inform the CPU that the transfer is complete.	

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Q.No.	Answer description	Marks Allotted
6 a)	SRAM (Static Random Access Memory) It is a type of semiconductor memory used to store binary data (0s and 1s). It is called static because it retains stored data as long as power is supplied without requiring periodic refreshing.	06 M
	Internal organization of an SRAM cell	
		- 03 M
	A typical SRAM cell contains the following components:	
	1) Two cross-coupled inverters • Form a bistable latch • Store one bit of data • Maintain either Logic 0 or Logic 1	
	2) Two access transistors • Connect the storage latch to the bit line • Controlled by the word line (WL) • Enable reading and writing operations.	

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Q.No.	Answer description	Marks Allotted
	3) Bit Lines (BL and $\overline{BL}$) • Two Complementary lines used for transferring data. • one carries the data and other carries its complement. 4) Word line (WL) • Select the SRAM cell for read or write operations. Working principle of SRAM • When one storage node contains 1, the other contains 0. • When one storage node contains 0, the other contains 1. • The feedback connection maintains the stored value. Read operation in SRAM 1) precharging: Both bit lines (BL and $\overline{BL}$) are precharged to logic High. 2) Word Line Activation: The word line is activated by setting WL = 1. 3) Connection: The access transistors turn ON and connect the internal storage nodes to the bit lines. 4) Data Transfer: Depending on the stored value, one bit line develops a small voltage drop.	02M

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Q.No.	Answer description	Marks Allotted
	5) Sensing: The Sense Amplifier detects the voltage difference between two bit lines. 6) Output: The Sense Amplifier produces the stored binary value at the output. Write operation in SRAM 1) Data Input: The data to be stored is supplied to the write circuit. 2) Bit Line Setup: The write driver places the data and its complement on BL and $\overline{BL}$ 3) Word Line Activation: WL is set to High 4) Access Transistors ON: The access transistors connect the bit lines to the internal storage nodes. 5) Data Storage: The write driver forces the latch into the required state. 6) Word Line Deactivation: WL is set to Low, disconnecting the cell from the bit line 7) Data Retention: The cross-coupled inverters maintain the newly written data as long as power is supplied.	02M

Course Teacher | Domain Expert | Chairperson

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